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METHOD OF STATEMENT FOR AIR CONDITIONING & MECHANICAL VENTILATION (ACMV) SYSTEM TESTING, COMMISSIONING, AND AIR BALANCING

CEILING CASSETTE & WALL MOUNTED UNIT (FCU) SYSTEM AIR BALANCING PROCEDURE (REF.PIPE SYSTEM)

Prepared BY: _____
Subcontractor (Name & Signature) (Date)

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Project(M&E) (Name & Signature) (Date)
In- Charge

Approval BY: _____
Consultant (Name & Signature) (Date)

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Objective:

This document provides a detailed, systematic procedure for the air balancing, testing, and commissioning of non-ducted VRF **Ceiling Cassette & Wall-Mounted Variable Refrigerant Flow (VRF)** or Direct Expansion (DX) Fan Coil Unit (FCU) systems. The primary goal is to provide auditable proof that the FCU, its associated refrigerant circuit, and the entire air distribution network are installed and performing in full accordance with the project design specifications, approved shop drawings,

To verify the entire VRF system is communicating and delivering cooling effectively to the indoor unit. This test is performed after the air-side verification is complete.

2.0 Scope of Work

This Method Statement covers all on-site activities required to achieve a fully functional and balanced system, including:

1. Verification of pre-requisite conditions, including the integrity of the refrigerant system.
2. Systematic air-side balancing of the non-ducted network FCU.
3. Functional testing and performance verification of the automated refrigerant control system.
4. Verification of local and central controller functions.
5. All associated safety procedures and documentation.

3.0 References

1. Project-Specific Approved Shop Drawings, Technical Specifications, and Equipment Schedules.
2. VRF System Manufacturer's Installation.
3. **Workplace Safety and Health Act (WSH Act)** and subsidiary legislation.

4.0 Responsibilities

1. **Project Manager:** Overall responsibility for the successful execution of the project, ensuring all resources are available.
2. **Site Engineer:** To provide technical guidance, review and approve the method statement and test reports. To witness and verify the testing and commissioning process.
3. **Commissioning Coordinator:** To plan, coordinate, and execute all testing and commissioning activities. To ensure that all procedures are followed and documented.
4. **ACMV Technicians:** To carry out the installation, pre-commissioning checks, and testing and commissioning works under the supervision of the Commissioning Site Engineer.

5. **BMS Controls Contractor:** To be present during the testing of control systems and to make any necessary adjustments to the BMS.
6. **Safety Personal:** To ensure that all on-site activities comply with the WSH Act and project-specific safety requirements.

5.0 Preparation / Equipment Required

1. Submission and approval of this method statement.
2. Conducting a toolbox meeting with all relevant personnel to discuss the scope of work, procedures, and safety precautions.
3. Verification that the ACMV system installation is complete and in accordance with the approved drawings.
4. Ensure the Calibrated air capture hood is assembled correctly with the appropriate skirt size (e.g., 600mm x 600mm or 1200 x600) that matches the nominal size of the diffuser being tested.
5. Calibrated Multi-function Meter (e.g., TSI VelociCalc 9600 series or equivalent) and the Vane Anemometer.
6. **Risk Assessment** for this activity has been approved, and a **Permit-to-Work (PTW)** has been issued if required.
7. Safe access equipment (e.g., ladder, mobile scaffold)
8. Appropriate hand tools (e.g., VCD adjustment tools, Allen keys).
9. Appropriate Personal Protective Equipment (PPE): safety glasses, gloves, safety footwear.
10. Calibrated testing equipment and air balancing test sheets are ready.

 Air Capture Hood Instruments	 Vane Anemometer	 Multi-function Anemometer
 Multi Clamp Meter	 PPE	 Ladder

(This section in the final report would include clear, dated photographs of the specific calibrated instruments used on-site, showing their serial numbers and calibration stickers.)

6.0 Pre-Requisites

1. **Refrigerant System Integrity:** The entire VRF/DX system, including all indoor and outdoor units, has been successfully **pressure tested with Oxygen-Free Nitrogen (OFN)**, **evacuated to a deep vacuum**, and **charged with the precise quantity of refrigerant** calculated by the manufacturer. This is a critical hold point.
2. **System Communication:** The system has been energized and a full **"auto-addressing" cycle** has been completed successfully, with the outdoor unit confirming communication with all connected indoor units and controllers.
3. **Physical Completion:** The FCU, filters, Copper pipe, powered on and condensate drain piping are fully installed, clean, and free of debris.
4. The Building Management System (BMS) is capable of commanding the FCU to start or Stop the FCU system from the starter CRC or Remote controller Manually

6.1 Refrigerant Performance Verification

Initiate Full Load Test from the central controller.

1. **Performance Data Collection:** Allow the unit to stabilize for 15-20 minutes. Measure the Return Air Temperature (air entering the unit) and Supply Air Temperature (air leaving the unit).
2. **System Diagnostics:** Verify that no error codes are displayed on any controller.

7.0 Step-by-Step Balancing Procedure

7.1 Phase A: Air-Side Balancing

1. **Set Operating Mode:** Using the unit's remote controller, turn the unit ON, select "Cool" or "Fan" mode, and set the fan speed to the level specified in the design schedule (e.g., High, Medium, Low).
2. **For Ceiling Cassette Units:** Using a calibrated **Air Capture Hood** with the correct skirt size (600x600 or 1200 x600), place the hood firmly against the ceiling, sealing around the cassette's fascia panel. The instrument will provide a direct, total volumetric airflow reading (CMH).
3. **For Wall-Mounted Units:** Use a calibrated **Vane Anemometer**. Perform a detailed traverse across the supply air louver, taking multiple velocity readings in a grid pattern. Calculate the average velocity (m/s).

B. Compare & Adjust:

4. **For Ceiling Cassette:** Compare the direct airflow reading from the hood with the manufacturer's technical data for that fan speed.
5. **For Wall-Mounted:** Calculate the airflow using the formula: **Airflow (CMH) = Average Velocity (m/s) × Louver Free Area × 3600**. Must be sourced from the Approval equipment drawing data. Compare this calculated airflow with the Approval equipment drawing.
6. **Adjustment:** If the measured/calculated airflow is outside the +/- 10% tolerance, the only adjustment is to select the next available fan speed on the remote controller that best matches the design requirement.
7. **Louver Function Test:** Activate the "Swing" function from the remote and verify that all directional louvers are moving smoothly through their full range of motion.
8. **Record Airflow Data:** Document the final fan speed setting and the verified airflow reading/calculation on the test sheet.

7.2 Room Temperature

1. **Setup Equipment:** Attach the appropriate temperature and RH probe to the multi-function meter.
2. **Position Probe:** Hold the probe in the centre of the room, away from direct sunlight, supply air diffusers, or heat-generating equipment to get a representative ambient reading.
3. **Record Readings:** Allow the readings to stabilize for a few minutes. Record the room temperature (°C) Verify these readings are within the comfort criteria specified in SS 553 and SS 554.

8.0 Types of Local Controllers:

1. **Wired Remote Controller:** A physical keypad mounted on the wall, connected via a low-voltage cable to the indoor unit.
2. **Wireless Infrared (IR) Remote Controller:** A handheld remote similar to a TV remote, which communicates with an IR receiver on the indoor unit's fascia panel.

9.0 Functional Control Test

1. **ON/OFF Control** 
Press the ON/OFF button. Verify the indoor unit starts and stops accordingly.
Check that the unit's operating turns on and off.
2. **Mode Selection**   
Cycle through all available operating modes (e.g., Cool, Heat, Fan, Dry, Auto).
Verify the controller's display and the indoor unit respond to each mode change.

3. **Fan Speed Control**

Cycle through all available fan speeds (e.g., Low, Medium, High, Auto). Listen for the audible change in the indoor unit's fan speed.

4. **Temperature Setpoint**

Adjust the temperature setpoint up and down. Verify the display reflects the changes.

5. **Louver Control** ↔

Test the louver swing function (if applicable). Press the button to start and stop the automatic oscillation of the louvers. Test any manual louver position settings.

10.0 Commissioning With BMS (If the Project Have BMS System)

1. **Point-to-Point Verification:** Verify all status points (Run, Fault/Trip) and command points (Start/Stop, Speed Control) from the LCP are correctly wired and appear on the BMS graphic.
2. **Remote Command Test:** Place the fan's LCP selector switch in "AUTO" mode.
 - Send a "Start" command from the BMS and verify the fan starts.
 - Send a "Stop" command and verify the fan stops.
3. **Status Feedback Test:**
 - While the fan is running, verify the "Run Status" on the BMS is "ON" or "Running."
 - Manually trip the motor's overload relay at the LCP. Verify the fan stops and the "Fault" or "Trip" status appears on the BMS. Reset the fault.

11.0 Documentation and Handover

1. Throughout the process, record all readings, observations, and test results on the Commissioning Test Sheet for the system.
2. Any failures, discrepancies, or non-conformities shall be recorded on a "Punch List" for rectification.
3. Upon successful completion of all tests, the Commissioning Engineer and Supervisor will sign the test sheet, confirming the System is safe and fully functional.
4. The completed and signed documentation will be included in the final Testing & Commissioning report for handover.
5. Compile the report for submission and verification with Maincon M&E / RTO or Consultant.

12.0 Health and Safety

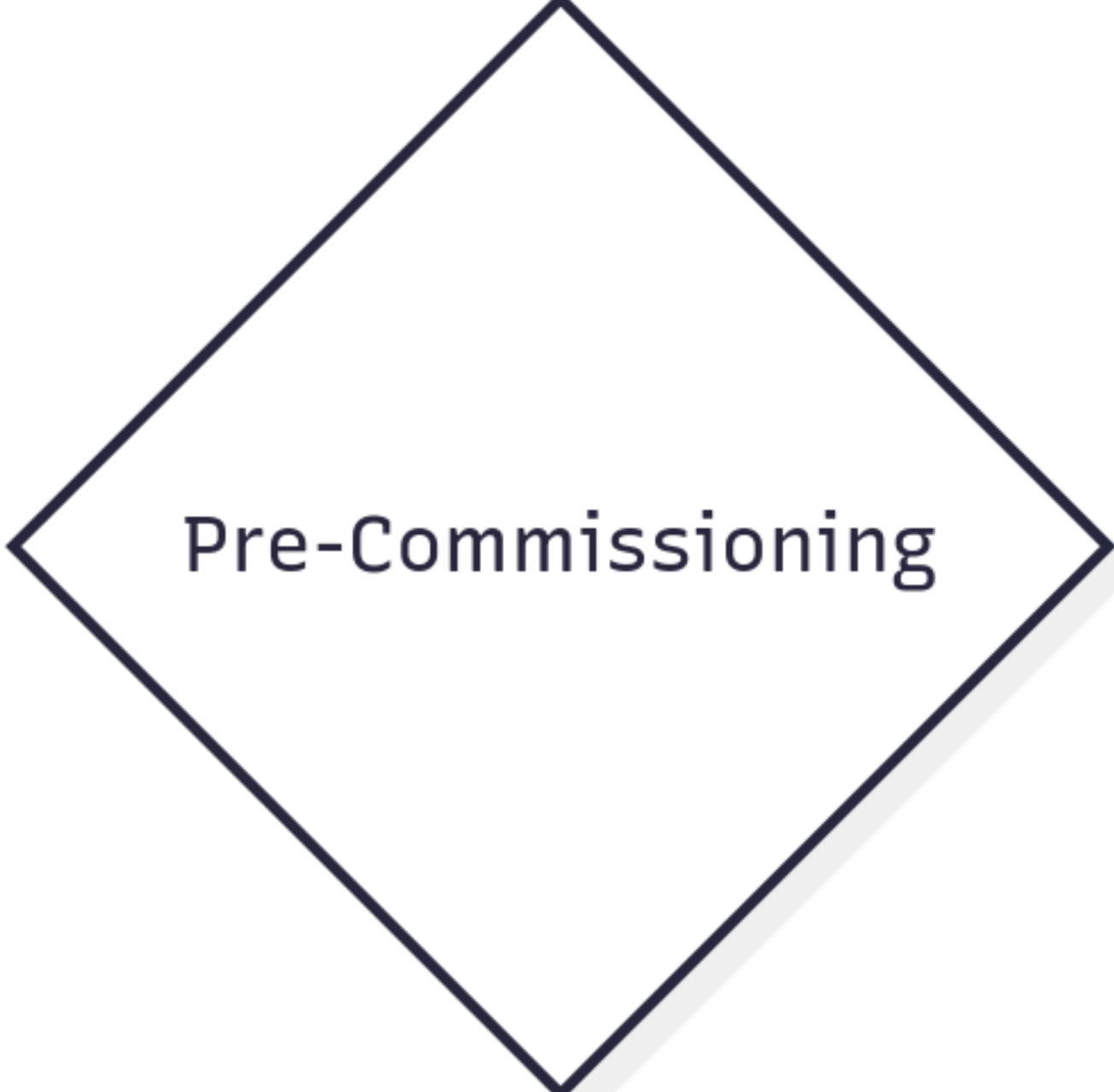
All work will be carried out in strict adherence to the project's Health, Safety, and Environment (HSE) plan and the WSH Act. Key safety measures include:

1. All personnel will wear the appropriate Personal Protective Equipment (PPE).
2. Lock-out/Tag-out (LOTO) procedures will be implemented before any work is carried out on energized equipment.
3. Safe access will be ensured for all work areas, especially at heights.
4. Good housekeeping will be maintained at all times.

13.0 APPENDIX

VRF FCU Testing Flowchart

Installation Complete



Check Piping, Wiring, Drains

Pressure & Leak Test Passed?

No

Yes

Rectify Leaks

Evacuate & Charge Refrigerant

Power Up & Initiate Test Run

Check Unit Communication

Test Each Indoor Unit - Cooling or Heating

Performance OK?

No

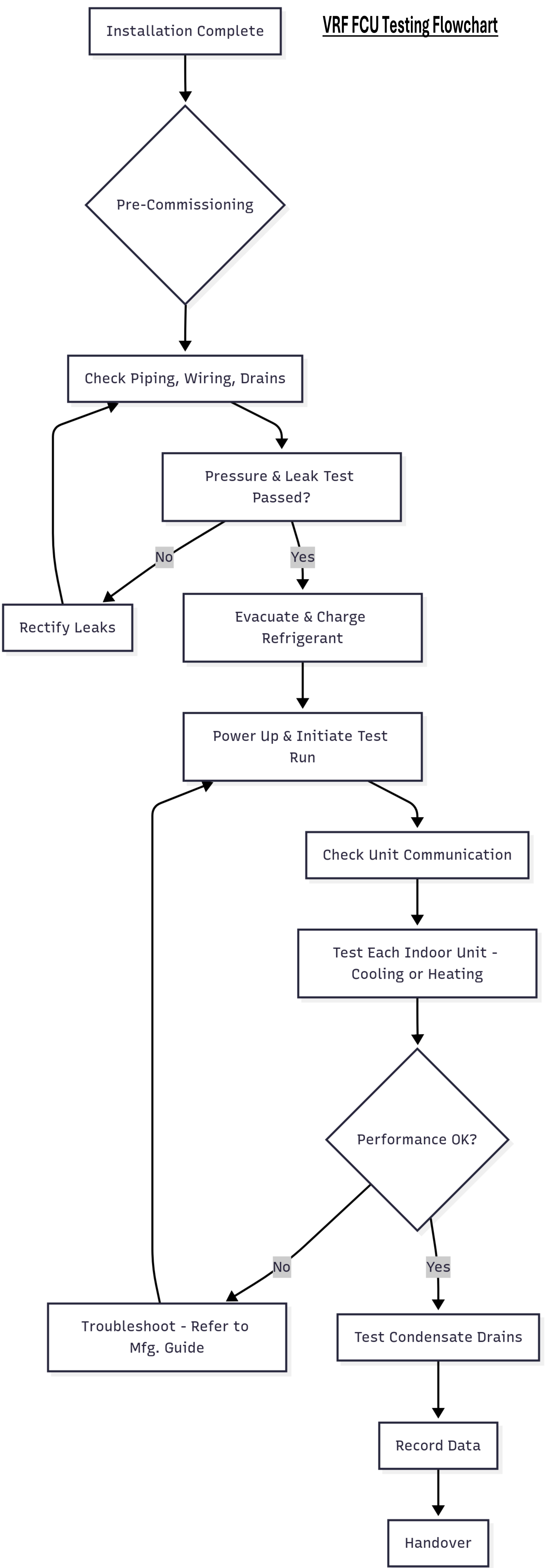
Yes

Troubleshoot - Refer to Mfg. Guide

Test Condensate Drains

Record Data

Handover



OPERATION DATA FOR INDOOR (SPLIT, VRV -AHU/FCU) UNIT			
PROJECT NAME			
LOCATION		INDOOR DESIGNATION	
AREA SERVED		INDOOR MODEL	
SYSTEM		MANUFACTURER	

S/No	DESCRIPTION	UNIT	TECHNICAL DATA
1	Total cooling capacity	kW	
2	Type of Refrigerant	-	
3	Motor Power	kW	
4	Power Supply	V/P/Hz	

S/No	COMPONENT	ITEM	UNIT	MEASURED DATA
1	Air on coil temperature (DB/WB)		°C	
2	Air off coil temperature (DB/WB)		°C	
3	System operating current	Starting current	Amp	
		Running current	Amp	
4	Supply air flow		CMH	
5	Return / Fresh Air Flow		CMH	
6	External static pressure		Pa	
7	Others			

Comments: (If there is no additional comments enter NONE)

Tested By		Witnessed By	Verified By
Signature			
Name			
Designation			
Company			
Date			

AIR BALANCING TEST REPORT

PROJECT NAME			
LOCATION		EQUIPMENT DESIGNATION	
AREA SERVED		EQUIPMENT MODEL	
SYSTEM		MANUFACTURER	

[illegible]

Comments: (If there is no additional comments enter NONE)

Tested By		Witnessed By	Verified By
Signature			
Name			
Designation			
Company			
Date			

**Air Flow Testing & Commissioning Report**

Project Title :	Power Source (V/Ph/Hz):
Equipment Model:	FORMULA:(CMH = Average Velocity × (Duct or Grille Size) ×3600 CMH
Starting Ampere (A):	Running Ampere (A)
Location:	

No:	Equipment Designation	Area Served	Design CMH	Louver / Duct / Grille Size	P-1	P-2	P-3	P-4	P-5	Average Vel.M/s	Total CMH

Comments :-		
Tested By :	Witnessed By:	Verified By :
Signature :	Signature :	Signature :
Name :	Name :	Name :
Designation :	Designation :	Designation :
Company :	Company :	Company :
Date:	Date:	Date:

TEMPERATURE MEASUREMENT REPORT			
PROJECT NAME			
LOCATION		EQUIPMENT DESIGNATION	
AREA SERVED		EQUIPMENT MODEL	
SYSTEM		MANUFACTURER	

S/No	Room Designation	Temperature (°C)		Remarks
		Design	Measured	
			Dry /Wet	
1			/	
2			/	
3			/	
4			/	
5			/	
6			/	
7			/	
8			/	
9			/	
10			/	
11			/	
12			/	
13			/	
14			/	
15			/	

Comments: (If there is no additional comments enter NONE)

Tested By		Witnessed By	Verified By
Signature			
Name			
Designation			
Company			
Date			



GRACECAL PTE LTD (GST Reg No: 201611630E)

31 Woodlands Close #02-05 Woodlands Horizon Singapore 737855
Tel: (65) 6742 3850 | Fax: (65) 6742 3849 | Email: sales@gracecal.com
Website: www.gracecal.com

CALIBRATION CERTIFICATE

CLIENT : **WINNER ENGINEERING PTE LTD**
18 Woodlands Industrial Park E1
Singapore 757738

CERTIFICATE NO : GCC241442
ISSUE DATE : 06-SEP-2024

INSTRUMENT : **MULTI FUNCTION ANEMOMETER**
MAKE : TSI
MODEL NO. : TA465-P-NB
PART NO. : ---
SERIAL NO. : TA4652029012

WORKS ORDER NO : GWO241601
DATE RECEIVED : 02-SEP-2024

DATE CALIBRATED : 06-SEP-2024
DUE DATE : 06-SEP-2025

AMBIENT TEMPERATURE:
(23 ± 5) °C
RELATIVE HUMIDITY:
(55 ± 10) % r.h.

RANGE : -----
GRADUATION : -----
(TAG NO.) : -----
PAGE : 1 of 2
STATUS : AS FOUND

The described instrument has been calibrated at Laboratory under the ambient conditions stated above.

This certificate provides traceability of measurement to the International System of Units (SI) and/or to units of measurement realized at the National Metrology Centre (NMC), Singapore or other recognized national metrology institutes.

The Instrument was calibrated by comparison with laboratory calibration testing equipment.

1. Gas Deadweight Pressure Tester S/N: 558735

Calibration Method

Calibration Method is in accordance with ISO/IEC 17025. The calibration method was carried out according to Technical Calibration Procedure AS-TM-M03 as a guide.

Calibration Results

1. The results of calibration are given on the attached calibration data sheet(s).
2. The expanded uncertainty of measurement associated with the calibration is estimated at a confidence level of approximately 95% with a coverage factor of k=2.00.
3. The user should determine the suitability of the instrument for its intended use.

Calibrated By:

PREM

ID: 748656

Approved By:

KUMAR

ID: 748670





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CALIBRATION CERTIFICATE

CERTIFICATE NO : GCC241442
ISSUED DATE : 06-SEP-2024

WORKS ORDER NO : GWO241601
PAGE : 2 of 2

APPLIED ACTUAL VALUE (Pa)			MEAN INSTRUMENT READING (Pa)			
	Before Adj.	Ascending After Adj.	Error	Before Adj.	Descending After Adj.	Error
Pressure Measurement						
0.00	0.1	---	-0.10	0.0	---	0.00
50.46	50.7	---	-0.24	50.7	---	-0.24
477.53	477.8	---	-0.27	477.8	---	-0.27
916.92	917.2	---	-0.28	917.2	---	-0.28